A Quadratic Field Criterion for Irrationality Exponent Two

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Abstract

Using the separated-weight interpolation theorems of Liu–Jiang–Zhang, we establish a quadratic field criterion for irrationality exponent two. Let D be a nonsquare integer and x a nonzero real number. If $\alpha = e^{2\sqrt{D}\,x}$ belongs to $\mathbb{Q}(\sqrt{D})$ and has norm one, then $\mu(x) = 2$. For $D < 0$ the norm condition is automatic; for $D > 0$ only the distinct-centre interpolation theorem is needed. The proof estimates one algebraic determinant in both embeddings and combines the estimates through its norm. Consequences include rational arctangents, quadratic multiples of π, and normalized logarithms of positive real quadratic numbers of norm ± 1.

1 Introduction

For an irrational real number x, its irrationality exponent $\mu(x)$ is the supremum of the real numbers $\nu > 0$ for which

$$0 < |x - p/q| < q^{-\nu}$$

holds for infinitely many coprime integers p, q with $q \geq 2$. We consider a criterion in which the exponential of a real number lies in a quadratic field. The geometric ingredients are the distinct-centre and bounded-fibre interpolation theorems of Liu–Jiang–Zhang [1, Theorems 2 and 9], recalled in Section 2. The determinant method follows [2, Sections 3–4].

Throughout, a nonsquare integer means an integer that is not the square of an integer; in particular, it is nonzero. For such an integer D, fix

$$\sqrt{D} = \begin{cases} \text{the positive square root of } D, & D > 0, \\ \mathrm{i}\sqrt{|D|}, & D < 0, \end{cases} \qquad K_D = \mathbb{Q}(\sqrt{D}).$$

Write N for the field norm from K_D to $\mathbb{Q}$.

Theorem 1 (Quadratic field criterion)**.** *Let D be a nonsquare integer and $x \in \mathbb{R} \setminus \{0\}$. If*

$$\alpha = e^{2\sqrt{D}\,x} \in K_D, \qquad \mathrm{N}(\alpha) = 1, \tag{1.1}$$

then x is irrational and $\mu(x) = 2$. More precisely, for every $\nu > 2$ there is $Q(x, D, \nu)$ such that

$$|x - p/q| > q^{-\nu} \qquad (p \in \mathbb{Z},\ q \in \mathbb{Z},\ q \geq Q(x, D, \nu)). \tag{1.2}$$

The fractions need not be reduced.

When $D < 0$, the nontrivial field automorphism is complex conjugation. Since $|e^{2\sqrt{D}\,x}| = 1$, membership in K_D already implies the norm condition in (1.1). Thus the theorem includes both real and imaginary quadratic fields. The bounds are pointwise in x and D; neither a uniform threshold nor a bound of the form $c(x)/q^2$ is asserted.

The common arithmetic step is to clear denominators in a nonzero determinant and take its field norm. Conjugation sends the two algebraic parameters $(\alpha, 2\sqrt{D})$ to $(\alpha^{-1}, -2\sqrt{D})$. This allows the same analytic estimate at the corresponding centres in both embeddings. Sections 3 and 4 establish these estimates, and Section 5 makes the parameter choices that prove the criterion. The applications in Section 6 include

$$\mu\Big(\arctan\frac{1}{n}\Big) = 2 \quad (n \geq 1), \qquad \mu\Big(\sqrt{\frac{a}{b}}\,\pi\Big) = 2 \quad (a, b \in \mathbb{Z}_{>0}),$$

as well as the real quadratic logarithm statement.

2 Separated-weight interpolation

All logarithms of positive real numbers are natural. The notation $\log(1+t)$ denotes its Taylor series at zero. Multi-index inequalities are componentwise, and $wa = \sum_i w_i a_i$.

For positive rational weights $W = (w_0, w_1, \ldots, w_m)$, let $\mathcal{P}_W(H)$ be the span over $\mathbb{C}$ of monomials $Y^h X^\ell$ with $w_0 h + w\ell \leq H$. For $V = (v_0, w_1/\theta, \ldots, w_m/\theta)$, let $\mathcal{J}_V(H)$ be the space of coefficients indexed by $t^s u^\beta$ with $v_0 s + w\beta/\theta < H$. Equivalently, it is the quotient of $\mathbb{C}[[t, u_1, \ldots, u_m]]$ by monomials of weight at least H.

Theorem 2 (Separated-weight interpolation [1, Theorems 2 and 9])**.** *Fix positive integers m, K, positive rational w_0, v_0, and rational $0 < \theta < 1$, with*

$$K(w_0/v_0)\theta^m < 1. \tag{2.1}$$

There are successive lower thresholds for positive rational weights $w_1, \ldots, w_m$, depending only on these fixed data and preceding weights, with the following properties.

(i) Distinct-centre statement. *For pairwise distinct $y_j \in \mathbb{C}^*$ and arbitrary $c_{ji} \in \mathbb{C}$, $0 \leq j < K$, the map*

$$\begin{aligned} \mathcal{P}_W(H) &\longrightarrow \bigoplus_{j=0}^{K-1} \mathcal{J}_V(H), \\ P &\longmapsto \big(P\big(y_j(1+t), (c_{ji} + u_i + \log(1+t))_{i=1}^m\big)\big)_j \end{aligned} \tag{2.2}$$

is onto for all sufficiently large $H = nR$.

(ii) Bounded-fibre statement. *If in addition $K\theta^m < 1$, the same conclusion holds for possibly repeated $y_j \in \mathbb{C}^*$, provided*

$$j \neq k, \quad y_j = y_k \quad \Longrightarrow \quad c_{ji} \neq c_{ki} \quad \text{for every } 1 \leq i \leq m.$$

In each statement, the positive integer R depends only on the weights; the threshold for n may depend on the centres. The successive weight thresholds are independent of the centres.

Part (i) is the statement used from [1, Theorem 2]. Part (ii) is [1, Theorem 9] with fibre-multiplicity parameter K. In the real quadratic case the multiplicative centres will be distinct. In the imaginary quadratic case the bounded-fibre statement also permits roots of unity, including $\alpha = 1$.

3 Conjugate determinants and the arithmetic estimate

Fix D, x, α as in Theorem 1. Write

$$\gamma = 2\sqrt{D}, \quad \omega = \gamma x, \quad \mathcal{O} = \mathbb{Z}[\sqrt{D}], \quad \sigma(a + b\sqrt{D}) = a - b\sqrt{D}.$$

Then $\sigma\gamma = -\gamma$ and $\sigma\alpha = \alpha^{-1}$. Choose an integer $s_0 \geq 1$ with $s_0\alpha \in \mathcal{O}$. Its conjugate is cleared by the same s_0. For $z = a + b\sqrt{D} \in \mathcal{O} \setminus \{0\}$,

$$|z\sigma z| = |a^2 - Db^2| \geq 1. \tag{3.1}$$

The nonsquare hypothesis ensures that the integer in (3.1) is nonzero. For $D < 0$ the two absolute values are equal, whereas for $D > 0$ their product is the required lower bound. Fix $\nu > 2$, and suppose that approximations

$$|x - p_i/q_i| \leq q_i^{-\nu}, \qquad p_i \in \mathbb{Z} \setminus \{0\}, \quad q_i \geq 2, \qquad w_i = \lceil \log q_i \rceil \tag{3.2}$$

have been chosen with weights satisfying Theorem 2. Their eventual selection is justified in Section 5. Use the centres

$$y_j = \alpha^j, \qquad c_{ji} = j\gamma p_i/q_i \quad (0 \leq j < K). \tag{3.3}$$

If $D > 0$, then $\alpha > 0$ and $\alpha \neq 1$, so the α^j are pairwise distinct. If $D < 0$, each additive coordinate $j\gamma p_i/q_i$ is distinct as j varies, since $p_i \neq 0$. Thus the applicable centre condition in Theorem 2 holds in either case. For the bounded-fibre statement we also impose $K\theta^m < 1$; Section 5 supplies this inequality.

Fix $F > 2/\theta$, and put

$$T_i = \left\lceil \frac{Fw_i}{v_0} \right\rceil, \qquad G_i(t) = \sum_{1 \leq k < T_i} \frac{(-1)^{k+1} t^k}{k}.$$

The difference $G_i(t) - \log(1+t)$ starts at weight $v_0 T_i \geq Fw_i > w_i/\theta$. Therefore replacing the logarithms by G_i in (2.2) is an invertible change of each finite coefficient packet: it changes u_i only by terms of strictly greater weight. Surjectivity is preserved.

Index columns by monomials $P = Y^h X^\ell$ with $w_0 h + w\ell \leq H$, where $w\ell = \sum_i w_i \ell_i$. Index rows by triples

$$\rho = (j, s, \beta), \qquad 0 \leq j < K, \quad v_0 s + w\beta/\theta < H.$$

The entry is

$$[t^s u^\beta] P(\alpha^j(1+t), (j\gamma p_i/q_i + G_i(t) + u_i)_i). \tag{3.4}$$

For each sufficiently large admissible H, choose a nonzero square minor Δ_H containing every row. Let its size be $M = M_H$, and define

$$\bar{b} = \frac{1}{MH} \sum_{\rho = (j,s,\beta)} w\beta, \qquad w_* = \min_i w_i.$$

Counting integer points in a fixed simplex gives

$$M \sim \frac{K\theta^m H^{m+1}}{(m+1)! v_0 \prod_i w_i}, \qquad 0 \leq \bar{b} \leq \theta. \tag{3.5}$$

Here, and throughout the determinant estimates, all data other than H are fixed before H tends to infinity.

Use these exact columns also for the conjugate determinant $\Delta_H^- := \sigma(\Delta_H)$, and write $\Delta_H^+ := \Delta_H$. Its matrix entries are

$$[t^s u^\beta] P(\alpha^{-j}(1+t), (-j\gamma p_i/q_i + G_i(t) + u_i)_i). \tag{3.6}$$

Both determinants are nonzero. The polynomials G_i have rational coefficients and are unchanged by σ. In particular, their signs must not be reversed. We apply σ only to algebraic entries; no extension of σ to the real number x or to ω is used.

Lemma 3 (Arithmetic estimate)**.** *Put* $\Lambda = 4\log 2$ *and*

$$E_{\mathrm{ar}} = \frac{\Lambda F m}{v_0} + \Lambda \sum_i \frac{1}{w_i} + \frac{\theta}{w_*} + \frac{(K-1)\log s_0}{w_0}. \tag{3.7}$$

Then

$$\frac{\log|\Delta_H^+| + \log|\Delta_H^-|}{2MH} \geq -(1-\bar{b}) - E_{\mathrm{ar}}. \tag{3.8}$$

Proof. Let $L_i = \mathrm{lcm}(1, \dots, T_i - 1)$, with the empty least common multiple equal to one, and let $D_H = \prod_i L_i^{\lfloor H/w_i \rfloor}$. Multiply column $Y^h X^\ell$ by $s_0^{(K-1)h} \prod_i q_i^{\ell_i}$, and row (j, s, β) by $\prod_i q_i^{-\beta_i}$. For $\ell \geq \beta$, the resulting entry is

$$s_0^{(K-1)h} \alpha^{jh} \binom{\ell}{\beta} [t^s](1+t)^h \prod_i \left(j\gamma p_i + q_i G_i(t)\right)^{\ell_i - \beta_i};$$

otherwise it is zero. The factor involving α is $(s_0\alpha)^{jh} s_0^{(K-1-j)h} \in \mathcal{O}$. Since $\gamma \in \mathcal{O}$, multiplying the entry by D_H puts it in $\mathcal{O}$. Let $S_H > 0$ be the total rational scaling factor, so the resulting nonzero determinant is $z_H = S_H \Delta_H^+ \in \mathcal{O}$ and $\sigma z_H = S_H \Delta_H^-$. Equation (3.1) gives $S_H^2 |\Delta_H^+ \Delta_H^-| \geq 1$. Consequently

$$\begin{aligned} \tfrac{1}{2}(\log|\Delta_H^+| + \log|\Delta_H^-|) \geq & -M \log D_H \\ & -(K-1)\log s_0 \sum_{\text{columns}} h \\ & - \sum_{\text{columns}} \sum_i \ell_i \log q_i + \sum_{\text{rows}} \sum_i \beta_i \log q_i. \end{aligned}$$

Use $h \leq H/w_0$, $\sum_i \ell_i \log q_i \leq H$, and

$$\sum_{\text{rows}} \sum_i \beta_i \log q_i \geq MH\bar{b} - MH\theta/w_*.$$

Finally, the elementary estimate $\log \mathrm{lcm}(1, \dots, N) \leq 4N \log 2$ implies

$$\frac{\log D_H}{H} \leq \Lambda \sum_i \frac{T_i}{w_i} \leq \frac{\Lambda F m}{v_0} + \Lambda \sum_i \frac{1}{w_i}.$$

For example, the least-common-multiple estimate follows by bounding each dyadic increment by the logarithm of the corresponding central binomial coefficient and summing. These inequalities give (3.8). □

4 Analytic estimates

4.1 Translation to entire functions

For each multi-index $a \in \mathbb{Z}^m_{\geq 0}$ and selected column P, set

$$f_{a,P}(z) = [u^a] P(e^z, z + u_1, \dots, z + u_m).$$

For $P = Y^h X^\ell$, this equals $\binom{\ell}{a} e^{hz} z^{|\ell| - |a|}$ if $a \leq \ell$, and zero otherwise. Thus only indices with $wa \leq H$ occur.

Lemma 4 (Exact translation with error bounds)**.** *For either sign $\varepsilon \in \{+1, -1\}$, every row of the corresponding matrix is a sum of at most*

$$Q_H = (\lfloor H \rfloor + 1)^{2m}(\lfloor H/v_0 \rfloor + 1)$$

scalar multiples of rows

$$\left([t^\ell] f_{a,P}(\varepsilon j\omega + \log(1+t))\right)_P, \qquad a \geq \beta, \quad wa \leq H, \quad 0 \leq \ell \leq s.$$

The scalar ξ belonging to such an index a satisfies

$$|\xi| \leq \exp\{-\nu w(a-\beta) + HE_{\mathrm{tr}}\}, \qquad E_{\mathrm{tr}} = \frac{\nu}{F} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(K|\gamma|) + \nu}{w_*}. \tag{4.1}$$

Proof. Write $\epsilon_i = \varepsilon j\gamma(p_i/q_i - x)$ and $\tau_i = G_i - \log(1+t)$. For the plus matrix use $\varepsilon = 1$; for the conjugate matrix use $\varepsilon = -1$. The identity $e^{\varepsilon j\omega} = \alpha^{\varepsilon j}$ makes the row expansion exact: with $z = \varepsilon j\omega + \log(1+t)$, its substituted coordinates are $Y = e^z$ and $X_i = z + u_i + \epsilon_i + \tau_i(t)$. Expanding in the u_i, then in $\epsilon_i + \tau_i$, gives

$$\mathrm{row}^{\varepsilon}_{j,s,\beta} = \sum_{\substack{a \geq \beta \\ wa \leq H}} \sum_{0 \leq e \leq a-\beta} \sum_{k=0}^{s} \binom{a}{\beta} \binom{a-\beta}{e} \epsilon^{a-\beta-e} \left([t^k] \prod_i \tau_i(t)^{e_i} \right) \cdot \left([t^{s-k}] f_{a,P}(\varepsilon j\omega + \log(1+t))\right)_P. \tag{4.2}$$

All the coefficients multiplying these rows are independent of the column P.

A nonzero tail coefficient requires $\sum_i e_i T_i \leq k \leq s$, so $we \leq v_0 s/F < H/F$. On $|t| = 1/2$, $|\tau_i(t)| \leq 1$; Cauchy's estimate bounds the tail coefficient by $2^k \leq 2^s$. Also

$$|\epsilon_i| \leq K|\gamma|e^{\nu} e^{-\nu w_i}, \qquad \binom{a}{\beta}\binom{a-\beta}{e} \leq 4^{|a|}, \qquad |a| \leq H/w_*.$$

Here $K|\gamma|e^{\nu} > 1$. The lost factor $e^{\nu we}$ is at most $e^{\nu H/F}$, giving (4.1). Counting the choices of a, e, k proves the stated bound on their number. □

Set

$$R_0 = 4K(1 + |\omega|), \qquad c = \frac{\log 2}{4}, \qquad E_{\mathrm{hol}} = \frac{R_0}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2R_0)}{w_*}. \tag{4.3}$$

The radius depends on x, D, K, but these are fixed in the degree limit.

4.2 Repeated Taylor indices

Lemma 5 (Repeated Taylor indices)**.** *Choose one translated row from each expansion in Lemma 4, omitting its scalar coefficient, and call the resulting square matrix T. If n_a of its rows have index a, then*

$$|\det T| \leq \exp\left\{ -c \sum_a n_a^2 + MH(E_{\mathrm{hol}} + \rho_H) \right\}, \tag{4.4}$$

where $\rho_H \to 0$ uniformly in the selected columns and row choices.

Proof. On $|z| \leq R_0$, all functions $f_{a,P}$ have modulus at most

$$\mathcal{D}_H = \exp\left\{ H\left(\frac{R_0}{w_0} + \frac{\log(2R_0)}{w_*} \right) \right\}.$$

Write $f_{a,P}(z) = \sum_{r\ge 0} c_{a,r,P}(z/R_0)^r$. Then $|c_{a,r,P}| \le \mathcal{D}_H$. On $|t| = 1/2$,

$$|\varepsilon j\omega + \log(1+t)| \le (K-1)|\omega| + \log 2 < R_0/2.$$

It follows that a coefficient test of order ℓ on $(z/R_0)^r$ has modulus at most $2^\ell 2^{-r}$.

Expand $\det T$ using these Taylor series. If two rows choose the same pair (a, r), their coefficient vectors are equal, so that term vanishes. Thus the Taylor degrees are distinct inside each group with fixed a, and every surviving term has $\sum r \ge \sum_a \binom{n_a}{2}$. Use half of the geometric decay to exploit this inequality, and sum the other half freely. Since $\sum \ell \le MH/v_0$, the result is

$$|\det T| \le M! \mathcal{D}_H^M \, 2^{MH/v_0} 2^{-\frac{1}{2}\sum_a \binom{n_a}{2}} (1 - 2^{-1/2})^{-M}.$$

This also proves absolute convergence of the expansion. Using $\sum_a n_a = M$ gives (4.4), for example with

$$\rho_H = \frac{\log M + \frac{1}{4}\log 2 - \log(1 - 2^{-1/2})}{H}.$$

By (3.5), this tends to zero. □

4.3 Comparison of the determinant estimates

Proposition 6 (The two alternatives)**.** *Let $0 < A < 1$ and $0 < \eta < 1$. Define*

$$g = \nu(A(1-\eta) - \theta) - (1-\theta), \qquad L = \frac{\eta^2 K \theta^m}{(m+1)v_0 A^m}, \tag{4.5}$$

and put $E = E_{\mathrm{ar}} + E_{\mathrm{tr}} + E_{\mathrm{hol}}$. The approximations (3.2) are impossible if

$$g > 0, \qquad E < g, \qquad cL > 1 + E. \tag{4.6}$$

Proof. Set

$$D_A(H) = \#\{a \in \mathbb{Z}^m_{\ge 0} : wa \le AH\}, \qquad L_H = \frac{\eta^2 M}{H D_A(H)}.$$

The simplex count gives $D_A(H) \sim (AH)^m/(m! \prod_i w_i)$, so $L_H \to L$.

Expand either Δ_H^ε by (4.2). For each selection of rows, the product of scalar coefficients has modulus at most

$$\exp\left\{-\nu \sum_\rho w a_\rho + \nu MH\bar{b} + MHE_{\mathrm{tr}}\right\}.$$

If at least ηM selected indices obey $wa \le AH$, then Cauchy–Schwarz gives $\sum_a n_a^2 \ge MHL_H$. The scalar main exponent is nonpositive, since $a_\rho \ge \beta_\rho$. The normalized exponent is therefore at most $-cL_H + E_{\mathrm{tr}} + E_{\mathrm{hol}} + \rho_H$.

Otherwise more than $(1-\eta)M$ indices have $wa > AH$. Discarding the nonpositive collision contribution gives a normalized exponent at most $-\nu(A(1-\eta) - \bar{b}) + E_{\mathrm{tr}} + E_{\mathrm{hol}} + \rho_H$. There are at most Q_H^M selections. Hence

$$\frac{\log|\Delta_H^\varepsilon|}{MH} \le E_{\mathrm{tr}} + E_{\mathrm{hol}} + o(1) + \max\{-cL_H, -\nu(A(1-\eta) - \bar{b})\}. \tag{4.7}$$

The added term $\log Q_H/H$ tends to zero, uniformly in the choices. For the second alternative, use

$$\nu(A(1-\eta) - \bar{b}) - (1 - \bar{b}) = \nu A(1-\eta) - 1 - (\nu - 1)\bar{b} \ge g,$$

because $\bar{b} \le \theta$. For the first, use $1 - \bar{b} \le 1$ and $L_H \to L$. Under (4.6), both alternatives in (4.7) are strictly smaller, for all sufficiently large admissible H, than the averaged lower bound (3.8). Each determinant obeys this same upper bound, so their average does too. This is the required contradiction. Notice that taking the norm doubles both the arithmetic cost and the analytic saving: it causes no loss of exponent. □

5 Parameter choices and proof of the criterion

Fix $\nu > 2$. Choose positive rational numbers θ, A, B, C such that

$$0 < \theta < A < B < 1, \quad \nu(A-\theta) > 1-\theta, \quad C > 1, \quad CB < 1, \quad B < C\theta < 1. \tag{5.1}$$

Here is an explicit existence argument. Choose a rational $b_0 \in (1/2, 1-1/\nu)$. For a sufficiently small positive rational δ, take $\theta = 1-\delta$, $A = 1 - b_0\delta$. Then

$$\nu(A-\theta) - (1-\theta) = (\nu(1-b_0) - 1)\delta > 0, \quad \theta - A^2 = (2b_0 - 1)\delta - b_0^2\delta^2 > 0.$$

Choose rational $C \in (A/\theta, 1/A)$, and then rational $B \in (A, \min\{1/C, C\theta\})$. These satisfy (5.1). Finally choose $\eta > 0$ small enough that g, as defined in (4.5), is positive.

Proof of Theorem 1. Suppose that $|x - p/q| \le q^{-\nu}$ holds for arbitrarily large integer denominators q. Put $\varepsilon_0 = \min\{g/2, 1/2\}$. Choose $F > 2/\theta$ large enough that $\nu/F < \varepsilon_0/3$. For integers m tending to infinity, consider

$$K = \lfloor C^m \rfloor, \qquad w_0 = B^{-m}, \qquad v_0 = 2K\theta^m w_0. \tag{5.2}$$

These are rational interpolation data satisfying

$$K\theta^m \le (C\theta)^m < 1, \qquad K(w_0/v_0)\theta^m = \tfrac{1}{2}.$$

Furthermore,

$$\frac{K}{w_0} \le (CB)^m \longrightarrow 0, \qquad v_0 \sim 2(C\theta/B)^m \longrightarrow \infty, \qquad \frac{m}{v_0} \longrightarrow 0, \tag{5.3}$$

whereas

$$L = \frac{\eta^2 (B/A)^m}{2(m+1)} \longrightarrow \infty. \tag{5.4}$$

Recall that s_0, ω, γ depend only on the fixed number x and on D. The total error in Proposition 6 is exactly

$$\begin{aligned} E = \frac{\nu}{F} + \frac{\Lambda F m + 2\log 2}{v_0} + \frac{(K-1)\log s_0 + 4K(1+|\omega|)}{w_0} + \Lambda \sum_i \frac{1}{w_i} \\ + \frac{\theta + \log 4 + \log(K|\gamma|) + \nu + \log(2R_0)}{w_*}. \end{aligned} \tag{5.5}$$

By (5.3)–(5.4), fix m large enough that the two middle terms involving v_0, w_0 have sum less than $\varepsilon_0/3$, and $cL > 2$. This fixes K, w_0, v_0, R_0.

Now choose m approximations from the assumed unbounded set of denominators, successively. First require all weights to be large enough that the last two terms of (5.5) have sum less than $\varepsilon_0/3$; the estimate $\sum_i 1/w_i \le m/w_*$ suffices. Then impose, at each step, the successive lower threshold in Theorem 2. Those thresholds do not depend on the centres, so this selection is legitimate. Since $x \neq 0$, require the denominators to be large enough that $p_i \neq 0$. The centre conditions were verified in Section 3: the powers α^j are distinct when $D > 0$, and the additive coordinates separate every repeated fibre when $D < 0$.

We now have $E < \varepsilon_0 \le g/2 < g$ and $cL > 2 > 1 + E$. Proposition 6 gives a contradiction. The limit in H is taken only after this finite list of approximations has been fixed. This proves the asserted eventual strict lower bound for each $\nu > 2$.

The argument allowed unreduced fractions and zero error. If x were rational, multiples of its denominator would give exact approximations with arbitrarily large denominators, a contradiction. Thus x is irrational. Dirichlet's theorem supplies infinitely many rational approximations of exponent arbitrarily close to two from below, so $\mu(x) \ge 2$. The bound just proved gives $\mu(x) \le 2$, completing the proof. □

6 Applications

Throughout this section, arctan is the real principal branch.

Corollary 7 (Rational arctangents)**.** *If $a \in \mathbb{Z} \setminus \{0\}$ and $b \in \mathbb{Z}_{>0}$, then*

$$\mu\left(\arctan \frac{a}{b}\right) = 2.$$

In particular, $\mu(\arctan(1/n)) = 2$ for every integer $n \geq 1$. Each of these numbers satisfies the eventual strict bound (1.2) for every $\nu > 2$.

Proof. Let $x = \arctan(a/b)$ and take $D = -1$. Then $x \neq 0$ and

$$e^{2\mathrm{i}x} = \frac{1 + \mathrm{i}a/b}{1 - \mathrm{i}a/b} = \frac{b^2 - a^2 + 2ab\mathrm{i}}{a^2 + b^2} \in \mathbb{Q}(\mathrm{i}). \tag{6.1}$$

Its norm is one, so Theorem 1 applies. The integer $s_0 = a^2 + b^2$ clears its denominator. In particular, for $a = 1$ and $b = n$, the additional arithmetic term is $(K-1)\log(n^2+1)/w_0$, which tends to zero in the parameter choice for each fixed n.

When $a = b = 1$, one has $x = \pi/4$ and $\alpha = \mathrm{i}$. Although the multiplicative centres repeat with period four, the additive coordinates separate them. Thus this case is covered by the same bounded-fibre argument. □

Corollary 8 (Quadratic multiples of π)**.** *For all positive integers a, b,*

$$\mu\left(\sqrt{\frac{a}{b}}\,\pi\right) = 2.$$

More generally, if $\lambda \in \mathbb{R} \setminus \{0\}$ and $\lambda^2 \in \mathbb{Q}$, then $\mu(\lambda\pi) = 2$. All these numbers satisfy (1.2) for every $\nu > 2$.

Proof. Take $D = -ab$ and $x = \sqrt{a/b}\,\pi$. Then

$$2\sqrt{D}\,x = 2\mathrm{i}\sqrt{ab}\sqrt{a/b}\,\pi = 2\pi\mathrm{i}a, \qquad e^{2\sqrt{D}\,x} = 1 \in K_D.$$

The norm is one and $x > 0$, so Theorem 1 applies. No squarefreeness or coprimality is required. Here $s_0 = 1$; all multiplicative centres are one, and their additive coordinates still separate them.

For the scalar formulation, write $\lambda^2 = a/b > 0$ with $a, b \in \mathbb{Z}_{>0}$. The positive choice is the first assertion. Changing the sign preserves the approximation errors after replacing p by $-p$. □

Corollary 9 (Normalized real quadratic logarithms)**.** *Let d be a positive nonsquare integer and let $\alpha \in \mathbb{Q}(\sqrt{d})$ be positive with $\alpha \neq 1$ and $\mathrm{N}(\alpha) \in \{1, -1\}$. Then*

$$\mu\left(\frac{\log \alpha}{\sqrt{d}}\right) = 2.$$

The number $\log\alpha/\sqrt{d}$ also satisfies the eventual strict bound (1.2) for every $\nu > 2$. No integrality or unit hypothesis is required.

Proof. Set $D = d$ and $x = \log\alpha/\sqrt{d} \neq 0$. Then

$$e^{2\sqrt{D}\,x} = \alpha^2 \in K_D, \qquad \mathrm{N}(\alpha^2) = 1.$$

The conclusion follows from Theorem 1. □

Remark 10. The normalization in Corollary 9 is part of the criterion. If one instead uses $x = \log \alpha$ and $\gamma = 1$ for a norm-one real quadratic α, conjugation leaves jp_i/q_i fixed while replacing α^j by α^{-j}. Its displacement from the matching analytic centre $-jx$ is $j(p_i/q_i + x)$, which tends to $2jx$ rather than zero. The present proof therefore does not establish $\mu(\log \alpha) = 2$ without the normalization. Likewise, Corollary 8 asserts the result for scalars whose square is rational, not for arbitrary algebraic scalars.

AI declaration

We received help from GPT-6 Astra to determine implications from $\mu(\pi) = 2$ and Liu–Jiang–Zhang's interpolation theorems, as well as to form the accompanying Lean repository (since we aren't very familiar with Lean). Still, we both independently derived the quadratic criterion to summarize these findings. We checked the Lean statements, and we are very confident that the formalized statement remains faithful to the original statement to be proven.

References

[1] Jingwen Liu, Kai Jiang, and Pingwen Zhang. Irrationality exponents of logarithms of positive rational numbers. https://arxiv.org/abs/2610.10192v2, 2026. arXiv:2610.10192v2, October 8. Theorems 2 and 9.

[2] OpenAI. The irrationality exponent of π is 2. Manuscript, revision adc7f1241b42, 2026. September 24, Sections 3–4.